Stop-Think-AutoRegress: Language Modeling with Latent Diffusion Planning

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Abstract

The Stop-Think-AutoRegress Language Diffusion Model (STAR-LDM) integrates latent diffusion planning with autoregressive generation. Unlike conventional autoregressive language models limited to token-by-token decisions, STAR-LDM incorporates a "thinking" phase that pauses generation to refine a semantic plan through diffusion before continuing. This enables global planning in continuous space prior to committing to discrete tokens. Evaluations show STAR-LDM significantly outperforms similar-sized models on language understanding benchmarks and achieves $> 70\%$ win rates in LLM-as-judge comparisons for narrative coherence and commonsense reasoning. The architecture also allows straightforward control through lightweight classifiers, enabling fine-grained steering of attributes without model retraining while maintaining better fluency-control trade-offs than specialized approaches. Our code is available at https://github.com/justinlovelace/STAR-LDM.

1 Introduction

Large Language Models (LLMs), typically based on the Transformer architecture (Vaswani et al., 2017), have demonstrated remarkable capabilities across a vast range of natural language tasks (Kojima et al., 2022; Brown et al., 2020). Their success largely stems from the autoregressive (AR) generation paradigm, where text is produced sequentially, predicting one token at a time conditioned on the preceding context. However, this strictly left-to-right, token-by-token generation process fundamentally differs from human writing practices. Human authors frequently pause, reflect, plan ahead, and revise their text to ensure global coherence, stylistic consistency, and adherence to overarching goals Hayes & Flower (1996). In contrast, standard autoregressive models make irrevocable, local decisions at each step, which inherently limits their capacity for long-range planning or dynamic control over the generation process (Liu et al., 2021b). Consequently, ensuring that generated text globally satisfies complex constraints, maintains a consistent persona, or avoids undesirable content purely through local token predictions remains a significant challenge, often necessitating computationally expensive fine-tuning or reinforcement learning procedures (Ziegler et al., 2019; Ouyang et al., 2022).

Recent advancements in generative modeling, particularly score-based diffusion models (Ho et al., 2020; Song et al., 2020), offer a fundamentally different paradigm. These models generate data through an iterative denoising process, beginning with pure noise and guided by a learned score function. Crucially, this score function can be readily modified during generation to incorporate external guidance signals from lightweight classifiers (Dhariwal & Nichol, 2021; Ho & Salimans, 2021), enabling flexible control over various attributes of the generated output (e.g., style, content). This inherent controllability, which can be effectively applied in continuous spaces such as image latents (Rombach et al., 2021) or text embeddings (Li et al., 2022; Lovelace et al., 2023), presents a particularly promising direction for enhancing control mechanisms in large language models.

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Inspired by the planning capabilities suggested by human writing and the controllability of diffusion models, we propose *Stop-Think-AutoRegress Language Diffusion Model (STAR-LDM)*. *STAR-LDM* is a novel, unified architecture that integrates latent diffusion planning directly into the autoregressive generation framework. It introduces a "thinking" phase where standard token generation is paused. During this phase, the model utilizes a diffusion process operating in a continuous sentence embedding space to refine a latent representation (a soft prompt) that encodes a plan for the subsequent text. This refined latent representation then guides the resumption of autoregressive token generation. By jointly training the autoregressive and diffusion components, STAR-LDM learns to perform global semantic planning in a continuous space *before* committing to discrete token choices, enabling more coherent and controllable generation.

Our contributions are as follows: (1) We propose STAR-LDM, a unified architecture that integrates latent diffusion planning within an autoregressive language model. (2) We demonstrate that STAR-LDM achieves improved language generation quality. (3) We show that our unified architecture leads to substantial improvements on standard language understanding benchmarks, indicating that the planning mechanism enhances, rather than hinders, core language capabilities. (4) We validate that STAR-LDM's integrated diffusion component enables efficient, plug-and-play control over text attributes (like sentiment and toxicity) using lightweight classifiers during inference, without requiring model retraining. STAR-LDM represents a step towards language models that can "stop and think" entirely in latent space, leading to higher-quality, more controllable, and potentially more reliable text generation.

2 Background

Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2020; Kingma & Gao, 2023), are a class of generative models that iteratively refine a sample of noise to a sample from some data distribution in a coarse-to-fine manner. Given clean data $\mathbf{x}_{\text{data}}$ from $q(\mathbf{x}_{\text{data}})$, diffusion models learn a model $p_\theta(\mathbf{x}_{\text{data}})$ approximating $q(\mathbf{x}_{\text{data}})$. Here, $\mathbf{x}_{\text{data}}$ could be images, audio, or continuous latent representations relevant to NLP (Li et al., 2022; Rombach et al., 2021; Lovelace et al., 2024), such as the sentence embeddings used in this work (Ni et al., 2022).

Forward process. Diffusion models define a forward noising process that transitions clean data $\mathbf{x}_{\text{data}}$ to noise over continuous time $t \in [0, 1]$. The noisy latent $\mathbf{z}_t$ at time t is given by $q(\mathbf{z}_t|\mathbf{x}_{\text{data}}) = \mathcal{N}(\mathbf{z}_t; \alpha_t \mathbf{x}_{\text{data}}, \sigma_t^2 \mathbf{I})$, which can be sampled as:

$$\mathbf{z}_t = \alpha_t \mathbf{x}_{\text{data}} + \sigma_t \boldsymbol{\epsilon}, \quad \text{where} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{1}$$

Using the variance-preserving formulation (Ho et al., 2020) ($\sigma_t^2 = 1 - \alpha_t^2$), the noise level is parameterized by the log Signal-to-Noise Ratio (SNR), $\lambda_t = \log(\alpha_t^2/\sigma_t^2)$. The schedule α_t decreases such that $\mathbf{z}_0 \approx \mathbf{x}_{\text{data}}$ and the distribution of $\mathbf{z}_1$ approaches a standard normal, i.e., $q(\mathbf{z}_1|\mathbf{x}_{\text{data}}) \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Generative model. The generative process reverses the forward process, starting from noise $\mathbf{z}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iteratively denoising it to produce a final sample $\mathbf{z}_0$ that approximates the data distribution $q(\mathbf{x}_{\text{data}})$. This requires the score function $\nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t)$, the gradient of the log-density of noisy data at time t. Diffusion models introduce a neural network $\mathbf{s}_\theta(\mathbf{z}_t, t)$ to approximate this score. Sampling involves iteratively applying the learned model $\mathbf{s}_\theta$ (or an equivalent parameterization, like noise prediction) to reverse the diffusion from $t = 1$ to $t = 0$, often using solvers like DDPM (Ho et al., 2020).

Training objective. The score network $\mathbf{s}_\theta$ can be learned via denoising score matching (Song & Ermon, 2019). An equivalent and common objective (Ho et al., 2020) trains a network $\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)$ to predict the noise $\boldsymbol{\epsilon}$ added to $\mathbf{x}_{\text{data}}$ to get $\mathbf{z}_t$, using a loss like:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, \mathbf{x}_{\text{data}} \sim q(\mathbf{x}_{\text{data}}), \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[w(\lambda_t) \cdot \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)\|_2^2 \right], \tag{2}$$

where $\mathbf{z}_t = \alpha_t \mathbf{x}_{\text{data}} + \sigma_t \boldsymbol{\epsilon}$ and $w(\lambda_t)$ is an SNR-dependent weighting term (Kingma et al., 2021). The score and noise predictions are related by $\mathbf{s}_\theta(\mathbf{z}_t, t) = -\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)/\sigma_t$.

Plug-and-play control. A key advantage of diffusion, as opposed to autoregressive, models is their controllability via plug-and-play guidance (Song et al., 2020; Dhariwal & Nichol, 2021). To generate samples conditioned on $\mathbf{y}$, the sampling process is guided by the conditional score $\nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t|\mathbf{y})$. Using Bayes' rule, this conditional score can be decomposed:

$$\nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t|\mathbf{y}) = \underbrace{\nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t)}_{\text{Unconditional Score}} + \underbrace{\nabla_{\mathbf{z}_t} \log p_t(\mathbf{y}|\mathbf{z}_t)}_{\text{Guidance Term}}. \tag{3}$$

The first term is the standard score approximated by the unconditional diffusion model $\mathbf{s}_\theta(\mathbf{z}_t, t)$. The second term guides the sampling towards latents $\mathbf{z}_t$ that are likely to produce the condition $\mathbf{y}$. This guidance term can be approximated using gradients from a *separate classifier* trained on noisy data, which is often called *classifier guidance* (Dhariwal & Nichol, 2021).

In practice, a guidance scale $s \geq 0$ is introduced to modulate the strength of the condition:

$$\hat{\mathbf{s}}(\mathbf{z}_t, t, \mathbf{y}) = \nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t) + s \cdot \nabla_{\mathbf{z}_t} \log p_t(\mathbf{y}|\mathbf{z}_t). \tag{4}$$

Setting $s = 0$ yields unconditional generation, while $s > 1$ increases the influence of $\mathbf{y}$. This provides flexible "plug-and-play" control without retraining the base model for each condition $\mathbf{y}$. This inherent controllability offers a promising avenue to address the control challenges of autoregressive language generation, which motivates our work.

3 Stop-Think-AutoRegress Language Diffusion Model

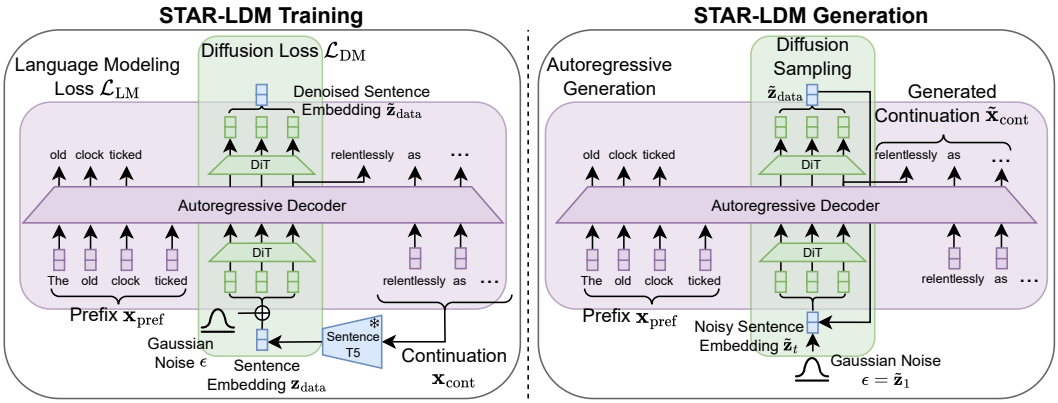

Figure 1: Unified diffusion-guided language model architecture illustrating the training (left) and generation (right) processes. See text for details.

We propose *Stop-Think-AutoRegress Language Diffusion Model (STAR-LDM)*, a unified architecture that augments autoregressive language modeling with latent diffusion planning. STAR-LDM can pause generation, plan in a continuous latent space, and resume autoregressive token generation with improved coherence.

Figure 1 depicts our proposed architecture, which consists of three major components, each with a primary role in enabling our "stop-think-autoregress" approach: (1) The **Autoregressive Decoder** (in purple) serves as the foundation of our model, handling token generation in a sequential manner similar to traditional language models. This component maintains the local fluency and grammatical structure necessary for coherent text. (2) The continuous **Sentence Embedding Space** (blue vectors) provides a latent planning space that captures semantic meaning. Using 768-dimensional Sentence-T5 XL embeddings (Ni et al., 2022), this space represents text at a higher level of abstraction than tokens, with representations that capture core semantic content while being invariant to shallow surface variations. (3) Two **Diffusion Transformers (DiTs)** (Peebles & Xie, 2022) (in green) facilitate the translation between the continuous embedding space and the discrete token representations used by the decoder. These transformer encoders are also conditioned on the noise level through adaptive normalization layers following Peebles & Xie (2022).